12.4.1 The freshness spectrum

Signal freshness falls along a spectrum:

- Embedded signals are fixed at the time of token issuance and valid until the token expires. They have zero runtime latency but can grow stale quickly.
- Cached signals are fetched periodically and reused for a set interval. They balance performance with acceptable staleness for data that rarely changes.
- Live signals are retrieved in real time for each decision, ensuring accuracy but adding latency and dependency on the upstream system.

Each signal source and delivery method defines its own point on this spectrum. Tokens from an identity provider might be trusted for an hour. HR data might be synced nightly. A risk engine might provide scores that expire in minutes.

12.4.2 Caching strategies

Caching is unavoidable in distributed authorization architectures. The key question is how long to cache and where.

- **At the PDP.** Many systems cache signal values or assembled entities within the PDP to avoid repeated lookups. This reduces latency but requires careful invalidation logic.
- **At the PIP.** Centralizing caches in PIPs lets multiple PDPs share consistent context, but increases the blast radius if the cache is stale.
- **At the edge or client.** Some applications cache tokens or context locally for offline or low-latency operation. This can be effective for low-risk data but introduces exposure if the device is compromised.

Short cache durations improve accuracy but increase load on upstream systems. Long durations improve performance, but risk outdated data. Systems often combine multiple cache tiers with differing time-to-live (TTL) values.